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1  PROGRAM PlantModel_ST
2  VAR
3      Gate1_Open_State : BOOL ;
4      Gate2_Open_State : BOOL ;
5      Gate3_Open_State : BOOL ;
6      Gate4_Open_State : BOOL ;
7      Gate1_Open_Timer : TON ;
8      Gate2_Open_Timer : TON ;
9      Gate3_Open_Timer : TON ;
10     Gate4_Open_Timer : TON ;
11     Gate1_Close_Timer : TON ;
12     Gate2_Close_Timer : TON ;
13     Gate3_Close_Timer : TON ;
14     Gate4_Close_Timer : TON ;
15     Mixer_Timer : TON ;
16     Init : BOOL := TRUE ;
17 END_VAR
18
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```
1  // Init - first cycle only
2  IF Init THEN
3      BinLevel_1 := 5000.0 ;
4      BinLevel_2 := 5000.0 ;
5      BinLevel_3 := 5000.0 ;
6      BinLevel_4 := 5000.0 ;
7      Weight_LoadCell := 0.0 ;
8      Init := FALSE ;
9  END_IF
10
11 // Gate travel delays
12 Gate1_Open_Timer ( IN := Gate_1 , PT := GATE_TRAVEL ) ;
13 Gate1_Open_State := Gate1_Open_Timer . Q ;
14 Gate1_Close_Timer ( IN := NOT Gate_1 , PT := GATE_TRAVEL ) ;
15 Gate1_Closed := Gate1_Close_Timer . Q ;
16
17 Gate2_Open_Timer ( IN := Gate_2 , PT := GATE_TRAVEL ) ;
18 Gate2_Open_State := Gate2_Open_Timer . Q ;
19 Gate2_Close_Timer ( IN := NOT Gate_2 , PT := GATE_TRAVEL ) ;
20 Gate2_Closed := Gate2_Close_Timer . Q ;
21
22 Gate3_Open_Timer ( IN := Gate_3 , PT := GATE_TRAVEL ) ;
23 Gate3_Open_State := Gate3_Open_Timer . Q ;
24 Gate3_Close_Timer ( IN := NOT Gate_3 , PT := GATE_TRAVEL ) ;
25 Gate3_Closed := Gate3_Close_Timer . Q ;
26
27 Gate4_Open_Timer ( IN := Gate_4 , PT := GATE_TRAVEL ) ;
28 Gate4_Open_State := Gate4_Open_Timer . Q ;
29 Gate4_Close_Timer ( IN := NOT Gate_4 , PT := GATE_TRAVEL ) ;
30 Gate4_Closed := Gate4_Close_Timer . Q ;
31
32 // Flow integration
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33  IF Gate1_Open_State THEN
34      Weight_LoadCell := Weight_LoadCell + FLOW_1 * CYCLE_TIME ;
35      BinLevel_1 := BinLevel_1 - FLOW_1 * CYCLE_TIME ;
36  END_IF
37  IF Gate2_Open_State THEN
38      Weight_LoadCell := Weight_LoadCell + FLOW_2 * CYCLE_TIME ;
39      BinLevel_2 := BinLevel_2 - FLOW_2 * CYCLE_TIME ;
40  END_IF
41  IF Gate3_Open_State THEN
42      Weight_LoadCell := Weight_LoadCell + FLOW_3 * CYCLE_TIME ;
43      BinLevel_3 := BinLevel_3 - FLOW_3 * CYCLE_TIME ;
44  END_IF
45  IF Gate4_Open_State THEN
46      Weight_LoadCell := Weight_LoadCell + FLOW_4 * CYCLE_TIME ;
47      BinLevel_4 := BinLevel_4 - FLOW_4 * CYCLE_TIME ;
48  END_IF
49
50  // Discharge
51  IF Discharge_Gate THEN
52      Weight_LoadCell := Weight_LoadCell - DISCHARGE_FLOW * CYCLE_TIME ;
53  END_IF
54
55  // Clamps (never negative)
56  Weight_LoadCell := MAX ( Weight_LoadCell , 0.0 ) ;
57  BinLevel_1 := MAX ( BinLevel_1 , 0.0 ) ;
58  BinLevel_2 := MAX ( BinLevel_2 , 0.0 ) ;
59  BinLevel_3 := MAX ( BinLevel_3 , 0.0 ) ;
60  BinLevel_4 := MAX ( BinLevel_4 , 0.0 ) ;
61
62  // Mixer
63  Mixer_Timer ( IN := Mixer_Run , PT := MIX_TIME ) ;
64  Mixer_Done := Mixer_Timer.Q ;
65
```